Testing lexical and pragmatical factors of 'conditional perfection' on its analyses

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Conditional perfection (CP) refers to the phenomenon where a bare conditional (e.g., *if p, q*) receives a bi-conditional (*iff*) interpretation [2]. The "invited" **Pragmatic Inferences (PIs)**, as shown in (1), range from (i) 'denying the antecedent' (not-P) or (ii) 'exhaustifivation' (only) to (iii) — a weaker version of conditional strengthening [4, 7, 8]. Furthermore, it has been shown that the extent of CP is subject to lexical and pragmatic factors: e.g., the **Focus Particle (FP)** is a CP-favoring factor, as *only if p, q* contains an overt exhaustifier (ii) and semantically encodes (i) [3, 1, 5]; conditional promises as in (1), whose antecendents lie in **Hearer Control (HC)** generate a higher degree of CP than those where the antecedent does not [6]. We predict that the lexical factor FP has a stronger effect than the pragmatic factor HC in generating CP, but how do these factors interact? Moreover, how do the three PIs compare with one another in relation to CP?

(1) **If** you mow the lawn, I'll give you five dollars. [***if p, q***]

- **(i)** If you **don't** mow the lawn, I **won't** give you five dollars. [***if*** $\neg$***p,*** $\neg$***q***]
- **(ii)** **Only if** you mow the lawn will I give you five dollars. [***only if p, q***]
- **(iii)** It's **not** the case that I will give you five dollars **anyway**. [$\neg$***(q no matter what)***]

Three experiments in German (ItemN/FillerN=12/24; SubjN=100/100/96 in **Exp1/2/3**, after outlier removal) were conducted using the same 2x2 factorial design (FP:*if/only if* x HC:nonHC/HC). The critical items, e.g., (2), each contained a context sentence (S1), a conditional sentence with/out FP or HC (S2), and (S3) a rating question on a 7-point scale (1-7: absolutely no-yes), which differed across experiments: *q if* $\neg$*p*? in Exp1, *q anyway*? in Exp2, *q only if p*? in Exp3, each targeting the PI **(i)/(iii)/(ii)**. Lower ratings in Exp1/2 but higher ratings in Exp3 are taken as indication of CP.

(2) **(S1-S2)** Ziad machte Schulaufgaben und sagte zu Paula: "**Wenn/Nur wenn** {es zum Thema einen Film gibt / du mir die Einleitung schreibst}, mache ich für dich das Poster." (*Z. did schoolwork and said to P.: "If/Only if there is a movie about the topic/you write the introduction for me, I'll make that poster for you."*) **(S3)** Was meinte Ziad: (*What did Ziad mean:*)
Exp1— Würde er das Poster für Paula machen, wenn es **keinen** Film zu dem Thema gibt/wenn Paula die Einleitung für ihn **nicht** schreibt? (*Would he make that poster if there is **no** movie about the topic/if P. does **not** write the introduction form him?*)
Exp2— Würde er das Poster **sowieso** machen? (*Would he make the poster **anyway**?*)
Exp3— Würde er für Paula das Poster machen, **nur** wenn es zu dem Thema einen Film gibt?/**nur** wenn Paula für ihn das Poster macht? (*Would he **only** make the poster for P. if there is a movie about the topic/ if she writes the introduction for him?*)

Results (cf. Fig 1) show: First, all experiments showed a main effect of FP ($p < 0.01$), with *nur wenn*$<_{ratings}$*wenn* in Exp1/2 and *nur wenn*$>_{ratings}$*wenn* in Exp3. Second, Exp2 showed a main effect of HC ($p = 0.03$), with HC$<_{ratings}$nonHC; this effect was not significant in Exp1/3. None of the studies found an interaction effect. Moreover, planned comparisons show that with *nur wenn*, the HC and nonHC conditions did not differ significantly in any of the experiments. With *wenn*, the HC and nonHC conditions differed significantly in Exp1/2 (HC$<_{ratings}$nonHC: $p < 0.01$), but not in Exp3. Altogether, our study provides convergent evidence for the CP-favoring effect of FP and HC, and shows a bigger CP-favoring effect of FP vs. HC, whose effect is only visible in absence of FP. We attribute the lack of a significant difference between *wenn*-HC and *wenn*-nonHC in Exp3 to a methodological deficit in that "nur" was present in both independent and dependent variables. In future work, we will focus on HC with additional other potential (dis)favoring factors for CP.

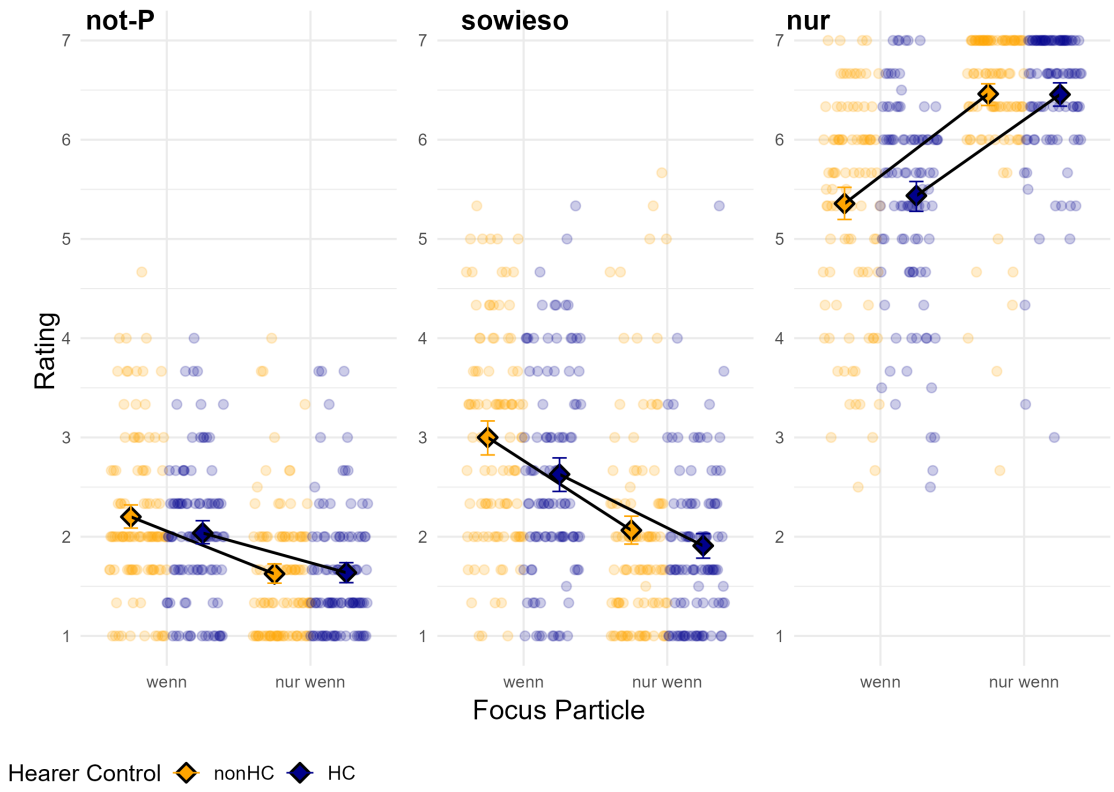

Figure 1: Grand means and subject means (squares vs. dots) of Exp1 (*not-P*), Exp2 (*sowieso* 'anyway') and Exp3 (*nur* 'only').

References

[1] Mathias Barthel, Rosario Tomasello, and Mingya Liu. "Conditionals in context: Brain signatures of prediction in discourse processing". In: *Cognition* 242 (2024).

[2] Michael L Geis and Arnold M Zwicky. "On invited inferences". In: *Linguistic Inquiry* 2.4 (1971).

[3] Elena Herburger. "Bare conditionals in the red". In: *Linguistics and Philosophy* 42.2 (2019).

[4] Laurence R. Horn. "From if to iff: Conditional perfection as pragmatic strengthening". In: *Journal of Pragmatics* 32.3 (2000).

[5] Mingya Liu and Mathias Barthel. "Semantics Processing of Conditional Connectives: German wenn 'if' Versus nur wenn 'only if'". In: *Journal of Psycholinguistic Research* 50 (2021).

[6] Ingrid van Canegem-Ardijns and William van Belle. "Conditionals and types of conditional perfection". In: *Journal of Pragmatics* 40.2 (2008).

[7] Johan van der Auwera. "Pragmatics in the last quarter century: The case of conditional perfection". In: *Journal of Pragmatics* 27.3 (1997).

[8] Kai von Fintel. "Conditional strengthening: A case study in Implicature". Unpublished ms. 2001.